

STAT 153 & 248 - Time Series

Lecture Thirteen

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Today, we shall wrap up our discussion of ridge regression and regularization from the frequentist (Cross-Validation) and Bayesian perspectives. Our model from the last few lectures is given by:

$$y_t = \beta_0 + \beta_1(t-1) + \beta_2\text{ReLU}(t-2) + \cdots + \beta_{n-1}\text{ReLU}(t-(n-1)) + \epsilon_t \quad (1)$$

where, as always, $\epsilon_t \stackrel{\text{i.i.d}}{\sim} N(0, \sigma^2)$. Here $\text{ReLU}(t-c) = (t-c)_+$ equals 0 if $t \leq c$ and equals $(t-c)$ if $t > c$.

The unknown parameters in this model are $\beta_0, \beta_1, \dots, \beta_{n-1}$ as well as σ .

Alternatively, (1) can be written as:

$$y = X\beta + \epsilon$$

where

$$X = \begin{pmatrix} 1 & 0 & 0 & \cdot & \cdot & \cdot & 0 \\ 1 & 1 & 0 & \cdot & \cdot & \cdot & 0 \\ 1 & 2 & 1 & \cdot & \cdot & \cdot & 0 \\ \cdot & \cdot & \cdot & \cdot & \cdot & \cdot & \cdot \\ \cdot & \cdot & \cdot & \cdot & \cdot & \cdot & \cdot \\ \cdot & \cdot & \cdot & \cdot & \cdot & \cdot & \cdot \\ 1 & n-1 & n-2 & \cdot & \cdot & \cdot & 1 \end{pmatrix} \text{ and } \beta = \begin{pmatrix} \beta_0 \\ \beta_1 \\ \beta_2 \\ \cdot \\ \cdot \\ \cdot \\ \beta_{n-1} \end{pmatrix}. \quad (2)$$

The ridge regression estimator $\hat{\beta}_{\text{ridge}}(\lambda)$ for β is given by the minimizer of:

$$\sum_{t=1}^n (y_t - \beta_0 - \beta_1(t-1) - \beta_2\text{ReLU}(t-2) - \cdots - \beta_{n-1}\text{ReLU}(t-(n-1)))^2 + \lambda(\beta_2^2 + \beta_3^2 + \cdots + \beta_{n-1}^2). \quad (3)$$

The objective function above can also be written as

$$L(\beta) := \|y - X\beta\|^2 + \lambda \sum_{j=2}^{n-1} \beta_j^2. \quad (4)$$

In the last lecture, we saw that

$$\hat{\beta}_{\text{ridge}}(\lambda) = (X^T X + \lambda J)^{-1} X^T y \quad (5)$$

which we proved via calculus equating the gradient $\nabla L(\beta)$ to zero. We can also prove this formula by just completing the square in the expression for $L(\beta)$. Specifically, we can write

$$L(\beta) = \|y - X\beta\|^2 + \lambda \sum_{j=2}^{n-1} \beta_j^2 = (y - X\beta)^T (y - X\beta) + \lambda \beta^T J \beta$$

where J is the $n \times n$ diagonal matrix whose diagonal entries are $0, 0, 1, \dots, 1$. In other words, the first two diagonal entries of J are 0 and the rest of the diagonal entries equal 1:

$$J = \begin{pmatrix} 0 & 0 & 0 & 0 & \cdot & \cdot & \cdot & 0 \\ 0 & 0 & 0 & 0 & \cdot & \cdot & \cdot & 0 \\ 0 & 0 & 1 & 0 & \cdot & \cdot & \cdot & 0 \\ 0 & 0 & 0 & 1 & \cdot & \cdot & \cdot & 0 \\ \cdot & \cdot & \cdot & \cdot & \cdot & \cdot & \cdot & \cdot \\ \cdot & \cdot & \cdot & \cdot & \cdot & \cdot & \cdot & \cdot \\ \cdot & \cdot & \cdot & \cdot & \cdot & \cdot & \cdot & \cdot \\ 0 & 0 & 0 & 0 & \cdot & \cdot & \cdot & 1 \end{pmatrix}$$

We now complete the square as:

$$\begin{aligned} L(\beta) &= (y - X\beta)^T (y - X\beta) + \lambda \beta^T J \beta \\ &= y^T y - 2y^T X\beta + \beta^T X^T X \beta + \lambda \beta^T J \beta \\ &= \beta^T (X^T X + \lambda J) \beta - 2y^T X \beta + y^T y \\ &= \left(\beta - \hat{\beta}_{\text{ridge}}(\lambda) \right)^T (X^T X + \lambda J) \left(\beta - \hat{\beta}_{\text{ridge}}(\lambda) \right) + y^T y - y^T X (X^T X + \lambda J)^{-1} X^T y \end{aligned}$$

where the last equality can be verified by direct calculation (just use $\hat{\beta}_{\text{ridge}}(\lambda) = (X^T X + \lambda J)^{-1} X^T y$ and expand the term $(\beta - \hat{\beta}_{\text{ridge}}(\lambda))^T (X^T X + \lambda J) (\beta - \hat{\beta}_{\text{ridge}}(\lambda))$). So the key equality is:

$$L(\beta) = \left(\beta - \hat{\beta}_{\text{ridge}}(\lambda) \right)^T (X^T X + \lambda J) \left(\beta - \hat{\beta}_{\text{ridge}}(\lambda) \right) + y^T y - y^T X (X^T X + \lambda J)^{-1} X^T y. \quad (6)$$

The first term above is a square term (because $X^T X + \lambda J$ is positive definite) and thus it is minimized when $\beta = \hat{\beta}_{\text{ridge}}(\lambda)$. This shows that $L(\beta)$ is minimized at $\hat{\beta}_{\text{ridge}}(\lambda)$.

1 Bayesian interpretation of ridge regularization

The starting point for Bayesian regularization is the observation that ridge regression coincides with posterior mean under the following prior: $\beta_0, \dots, \beta_{n-1}$ are all independent with

$$\beta_0, \beta_1 \stackrel{\text{i.i.d.}}{\sim} \text{unif}(-C, C) \text{ or } N(0, C) \quad \text{and} \quad \beta_2, \dots, \beta_{n-1} \stackrel{\text{i.i.d.}}{\sim} N\left(0, \frac{\sigma^2}{\lambda}\right) \quad (7)$$

Here $C \rightarrow \infty$ and σ, λ are fixed constants. Below write $|\sigma, \lambda$ to emphasize that σ and λ are being treated as fixed constants, so that we are effectively conditioning on σ and λ .

To see this, we first write the prior density as:

$$f_{\beta_0, \dots, \beta_{n-1} | \sigma, \lambda}(\beta_0, \dots, \beta_{n-1}) = f_{\beta_0}(\beta_0) f_{\beta_1}(\beta_1) \prod_{j=2}^{n-1} \frac{\sqrt{\lambda}}{\sqrt{2\pi}\sigma} \exp\left(-\frac{\lambda \beta_j^2}{2\sigma^2}\right) \quad (8)$$

For $f_{\beta_0}(\beta_0)$, if we are using $\beta_0 \sim \text{unif}(-C, C)$, then

$$f_{\beta_0}(\beta_0) = \frac{I\{-C < \beta_0 < C\}}{2C} \propto I\{-C < \beta_0 < C\}.$$

We then let $C \rightarrow \infty$ and replace the indicator by simply 1, leading to $f_{\beta_0}(\beta_0) \propto 1$. Similarly $f_{\beta_1}(\beta_1) \propto 1$. If we are instead using $\beta_0 \sim N(0, C)$, then

$$f_{\beta_0}(\beta_0) = \frac{1}{\sqrt{2\pi C}} \exp\left(-\frac{\beta_0^2}{2C}\right) \propto \exp\left(-\frac{\beta_0^2}{2C}\right).$$

We again let $C \rightarrow \infty$ and note that $\exp\left(-\frac{\beta_0^2}{2C}\right)$ converges to 1 as $C \rightarrow \infty$. Thus again $f_{\beta_0}(\beta_0) \propto 1$ and similarly $f_{\beta_1}(\beta_1) \propto 1$.

Replacing $f_{\beta_0}(\beta_0)$ and $f_{\beta_1}(\beta_1)$ both by 1 in (8), we obtain

$$f_{\beta_0, \dots, \beta_{n-1} | \sigma, \lambda}(\beta_0, \dots, \beta_{n-1}) \propto \lambda^{(n-2)/2} \sigma^{-(n-2)} \exp\left(-\frac{\lambda \sum_{j=2}^{n-1} \beta_j^2}{2\sigma^2}\right). \quad (9)$$

We combine this prior density with the usual linear regression likelihood given by:

$$\text{likelihood} \propto \frac{1}{\sigma^n} \exp\left(-\frac{\|y - X\beta\|^2}{2\sigma^2}\right). \quad (10)$$

This gives the posterior of $\beta_0, \dots, \beta_{n-1}$ as:

$$\begin{aligned} \text{posterior}(\beta_0, \dots, \beta_{n-1}) &= f_{\beta_0, \dots, \beta_{n-1} | \text{data}, \sigma, \lambda}(\beta_0, \dots, \beta_{n-1}) \\ &\propto \text{prior}(\beta_0, \dots, \beta_{n-1}) \times \text{likelihood} \\ &\propto \lambda^{(n-2)/2} \sigma^{-(n-2)} \exp\left(-\frac{\lambda \sum_{j=2}^{n-1} \beta_j^2}{2\sigma^2}\right) \frac{1}{\sigma^n} \exp\left(-\frac{\|y - X\beta\|^2}{2\sigma^2}\right) \\ &= \lambda^{(n-2)/2} \sigma^{-2n+2} \exp\left(-\frac{\|y - X\beta\|^2 + \lambda \sum_{j=2}^{n-1} \beta_j^2}{2\sigma^2}\right) \\ &= \lambda^{(n-2)/2} \sigma^{-2n+2} \exp\left(-\frac{L(\beta)}{2\sigma^2}\right) \end{aligned}$$

where $L(\beta)$ is the ridge regression objective function given in (4). Using the formula (6) for $L(\beta)$, we obtain

$$\begin{aligned} f_{\beta_0, \dots, \beta_{n-1} | \text{data}, \sigma, \lambda}(\beta_0, \dots, \beta_{n-1}) &= \lambda^{(n-2)/2} \sigma^{-2n+2} \exp\left(-\frac{(\beta - \hat{\beta}_{\text{ridge}}(\lambda))^T (X^T X + \lambda J) (\beta - \hat{\beta}_{\text{ridge}}(\lambda))}{2\sigma^2}\right) \\ &\quad \times \exp\left(-\frac{y^T y - y^T X (X^T X + \lambda J)^{-1} X^T y}{2\sigma^2}\right). \end{aligned}$$

Because σ and λ are being treated here as constants, we get

$$f_{\beta_0, \dots, \beta_{n-1} | \text{data}, \sigma, \lambda}(\beta_0, \dots, \beta_{n-1}) \propto \exp\left(-\frac{(\beta - \hat{\beta}_{\text{ridge}}(\lambda))^T (X^T X + \lambda J) (\beta - \hat{\beta}_{\text{ridge}}(\lambda))}{2\sigma^2}\right). \quad (11)$$

The right hand side above is the density (without the normalizing constant) of a multivariate normal density. The multivariate normal density $\beta \sim N(\mu, \Sigma)$ is given by

$$\frac{(2\pi)^{-n/2}}{\sqrt{\det \Sigma}} \exp\left(-\frac{(\beta - \mu)^T \Sigma^{-1} (\beta - \mu)}{2}\right) \propto \exp\left(-\frac{(\beta - \mu)^T \Sigma^{-1} (\beta - \mu)}{2}\right)$$

Comparing the above to (11), it is clear that

$$\mu = \hat{\beta}_{\text{ridge}}(\lambda) \quad \text{and} \quad \Sigma^{-1} = \frac{X^T X + \lambda J}{\sigma^2} \implies \Sigma = \sigma^2 (X^T X + \lambda J)^{-1}.$$

We have thus proved that the posterior corresponding to the prior (7) is given by

$$\beta \mid \text{data}, \sigma, \lambda \sim N(\hat{\beta}_{\text{ridge}}(\lambda), \sigma^2 (X^T X + \lambda J)^{-1}).$$

Therefore the ridge regression estimator is the posterior mean corresponding to the prior (7).

2 Bayesian inference for unknown λ

A natural way of dealing with unknown λ (and unknown σ) is to supplement the prior (7) with the following priors for σ and λ :

$$\log \sigma \sim \text{unif}(-C, C) \quad \text{and} \quad \lambda \sim f_\lambda.$$

Here again $C \rightarrow \infty$. For f_λ , we first find two values $\lambda_{\min}$ and $\lambda_{\max}$ such that the ridge regression estimate for $\lambda = \lambda_{\min}$ nearly overfits the data, and the ridge regression estimate for $\lambda = \lambda_{\max}$ is the linear regression line. We then take

$$\log \lambda \sim \text{unif}(\log \lambda_{\min}, \log \lambda_{\max})$$

which implies that

$$f_\lambda(\lambda) = f_{\log \lambda}(\log \lambda) \frac{1}{\lambda} = \frac{I\{\lambda_{\min} \leq \lambda \leq \lambda_{\max}\}}{\log \lambda_{\max} - \log \lambda_{\min}} \frac{1}{\lambda} \propto \frac{I\{\lambda_{\min} \leq \lambda \leq \lambda_{\max}\}}{\lambda}. \quad (12)$$

Combining these priors for σ and λ with (9), we obtain

$$f_{\beta_0, \dots, \beta_{n-1}, \sigma, \lambda}(\beta_0, \dots, \beta_{n-1}, \sigma, \lambda) \propto \lambda^{(n-2)/2} \sigma^{-(n-2)} \exp\left(-\frac{\lambda \sum_{j=2}^{n-1} \beta_j^2}{2\sigma^2}\right) f_\sigma(\sigma) f_\lambda(\lambda).$$

Using $f_\sigma(\sigma) = f_{\log \sigma}(\log \sigma) \frac{1}{\sigma} = I\{-C < \log \sigma < C\} / (2C\sigma) \propto I\{\sigma > 0\} / \sigma$ (assuming $C \rightarrow \infty$ so that the indicator can be replaced by $I\{\sigma > 0\}$), we get

$$f_{\beta_0, \dots, \beta_{n-1}, \sigma, \lambda}(\beta_0, \dots, \beta_{n-1}, \sigma, \lambda) \propto I\{\sigma > 0\} \lambda^{(n-2)/2} \sigma^{-n+1} \exp\left(-\frac{\lambda \sum_{j=2}^{n-1} \beta_j^2}{2\sigma^2}\right) f_\lambda(\lambda)$$

where $f_\lambda(\lambda)$ is (12).

Combining this prior with the likelihood (10), we get

$$f_{\beta, \sigma, \lambda \mid \text{data}}(\beta, \sigma, \lambda) \propto I\{\sigma > 0\} \lambda^{(n-2)/2} \sigma^{-2n+1} \exp\left(-\frac{L(\beta)}{2\sigma^2}\right) f_\lambda(\lambda).$$

Using the formula (6) for $L(\beta)$, we obtain

$$\begin{aligned} f_{\beta, \sigma, \lambda \mid \text{data}}(\beta, \sigma, \lambda) &= \lambda^{(n-2)/2} \sigma^{-2n+1} \exp\left(-\frac{(\beta - \hat{\beta}_{\text{ridge}}(\lambda))^T (X^T X + \lambda J) (\beta - \hat{\beta}_{\text{ridge}}(\lambda))}{2\sigma^2}\right) \\ &\quad \times \exp\left(-\frac{y^T y - y^T X (X^T X + \lambda J)^{-1} X^T y}{2\sigma^2}\right) f_\lambda(\lambda) I\{\sigma > 0\} \end{aligned}$$

As in the previous section, this implies

$$\beta \mid \text{data}, \sigma, \lambda \sim N(\hat{\beta}_{\text{ridge}}(\lambda), \sigma^2(X^T X + \lambda J)^{-1}).$$

We can also integrate out β to obtain the posterior for σ and λ . For this, we use the following formula for the normalizing constant of a multivariate normal density:

$$\begin{aligned} & \int \exp\left(-\frac{(\beta - \hat{\beta}_{\text{ridge}}(\lambda))^T (X^T X + \lambda J)(\beta - \hat{\beta}_{\text{ridge}}(\lambda))}{2\sigma^2}\right) d\beta \\ &= (2\pi)^{n/2} \sqrt{\det(\sigma^2(X^T X + \lambda J)^{-1})} \propto \sqrt{\det(\sigma^2(X^T X + \lambda J)^{-1})} = \sigma^n |X^T X + \lambda J|^{-1/2} \end{aligned}$$

where $|X^T X + \lambda J|$ is just another notation for the determinant of $X^T X + \lambda J$.

We thus obtain

$$\begin{aligned} & f_{\sigma, \lambda \mid \text{data}}(\sigma, \lambda) \\ &= \int f_{\beta, \sigma, \lambda \mid \text{data}}(\beta, \sigma, \lambda) d\beta \\ &\propto \lambda^{(n-2)/2} \sigma^{-n+1} |X^T X + \lambda J|^{-1/2} \exp\left(-\frac{y^T y - y^T X (X^T X + \lambda J)^{-1} X^T y}{2\sigma^2}\right) f_\lambda(\lambda) I\{\sigma > 0\}. \end{aligned}$$

This gives the joint posterior of σ and λ . We can now integrate over σ to deduce the marginal posterior of λ . This posterior can be used to figure out suitable values of λ (e.g., posterior mean or median).

$$\begin{aligned} f_{\lambda \mid \text{data}}(\lambda) &\propto f_\lambda(\lambda) \int_0^\infty \sigma^{-n+1} \lambda^{(n-2)/2} |X^T X + \lambda J|^{-1/2} \exp\left(-\frac{y^T y - y^T X (X^T X + \lambda J)^{-1} X^T y}{2\sigma^2}\right) d\sigma \\ &= f_\lambda(\lambda) \lambda^{(n-2)/2} |X^T X + \lambda J|^{-1/2} \int_0^\infty \sigma^{-n+1} \exp\left(-\frac{y^T y - y^T X (X^T X + \lambda J)^{-1} X^T y}{2\sigma^2}\right) d\sigma. \end{aligned}$$

Letting

$$A := y^T y - y^T X (X^T X + \lambda J)^{-1} X^T y,$$

we get

$$f_{\lambda \mid \text{data}}(\lambda) \propto f_\lambda(\lambda) \lambda^{(n-2)/2} |X^T X + \lambda J|^{-1/2} \int_0^\infty \sigma^{-n+1} \exp\left(-\frac{A}{2\sigma^2}\right) d\sigma.$$

By the change of variable $\sigma = s\sqrt{A}$, we obtain

$$\begin{aligned} f_{\lambda \mid \text{data}}(\lambda) &\propto f_\lambda(\lambda) \lambda^{(n-2)/2} |X^T X + \lambda J|^{-1/2} A^{-(n/2)+1} \int_0^\infty s^{-n+1} \exp\left(-\frac{1}{2s^2}\right) ds \\ &\propto f_\lambda(\lambda) \lambda^{(n-2)/2} |X^T X + \lambda J|^{-1/2} A^{-(n/2)+1} \\ &= \frac{f_\lambda(\lambda) \lambda^{(n-2)/2} |X^T X + \lambda J|^{-1/2}}{(y^T y - y^T X (X^T X + \lambda J)^{-1} X^T y)^{(n/2)-1}}. \end{aligned}$$

Thus

$$\begin{aligned} f_{\lambda \mid \text{data}}(\lambda) &\propto f_\lambda(\lambda) \frac{\lambda^{(n-2)/2} |X^T X + \lambda J|^{-1/2}}{(y^T y - y^T X (X^T X + \lambda J)^{-1} X^T y)^{(n/2)-1}} \\ &\propto I\{\lambda_{\min} \leq \lambda \leq \lambda_{\max}\} \frac{\lambda^{(n-4)/2} |X^T X + \lambda J|^{-1/2}}{(y^T y - y^T X (X^T X + \lambda J)^{-1} X^T y)^{(n/2)-1}} \end{aligned}$$

For computational stability, it is better here to work the posterior density of $\log \lambda$ instead of λ :

$$f_{\log \lambda | \text{data}}(x) = e^x f_{\lambda | \text{data}}(e^x) \\ \propto I\{\log \lambda_{\min} \leq x \leq \log \lambda_{\max}\} \frac{e^{x(n-2)/2} |X^T X + e^x J|^{-1/2}}{(y^T y - y^T X (X^T X + e^x J)^{-1} X^T y)^{(n/2)-1}} \quad (13)$$

In practice, we approximate this density by evaluating it at a grid of points x in the range $[\log \lambda_{\min}, \log \lambda_{\max}]$. One can then calculate the posterior mean or median of λ via the discrete approximation. This gives an alternative way of selecting λ that does not involve any kind of cross-validation (all calculations are done on the full dataset). Once λ is selected, the usual ridge regression estimator with the chosen value of λ can be used.

Since we are working with a full Bayesian prior for β, σ, λ , we can also do full Bayesian inference on all these parameters. The simplest way of doing this is via drawing posterior samples of $\beta^{(i)}, \sigma^{(i)}, \lambda^{(i)}$ of β, σ, λ for $i = 1, \dots, N$ (for a large N) in the following way:

1. First draw samples $\lambda^{(1)}, \dots, \lambda^{(N)}$ using the discrete approximation to the posterior (13).
2. For each $\lambda = \lambda^{(i)}$, we draw a sample $\sigma^{(i)}$ from the conditional posterior:

$$\sigma \mid \lambda = \lambda^{(i)}, \text{data}.$$

To actually do this, note from the above formula for $f_{\sigma, \lambda | \text{data}}(\sigma, \lambda)$, we can write

$$f_{\sigma | \lambda, \text{data}}(\sigma) \propto \sigma^{-n+1} \exp \left(-\frac{y^T y - y^T X (X^T X + \lambda J)^{-1} X^T y}{2\sigma^2} \right) I\{\sigma > 0\}.$$

The right hand side above corresponds to an inverse-gamma distribution for σ^2 . This can be seen by converting it into the density of $1/\sigma^2$:

$$f_{1/\sigma^2 | \text{data}, \lambda}(x) \propto f_{\sigma | \text{data}, \lambda}(x^{-1/2}) x^{-3/2} \\ \propto I\{x > 0\} (x^{-1/2})^{-n+1} \exp \left(-x \frac{y^T y - y^T X (X^T X + \lambda J)^{-1} X^T y}{2} \right) x^{-3/2} \\ = I\{x > 0\} x^{(n-4)/2} \exp \left(-x \frac{y^T y - y^T X (X^T X + \lambda J)^{-1} X^T y}{2} \right).$$

Thus

$$\frac{1}{\sigma^2} \mid \text{data}, \lambda \sim \text{Gamma} \left(\frac{n}{2} - 1, \frac{y^T y - y^T X (X^T X + \lambda J)^{-1} X^T y}{2} \right). \quad (14)$$

Thus $\sigma^{(i)}$ can be chosen as

$$\sigma^{(i)} = G_i^{-1/2} \quad \text{where } G_i \sim \text{Gamma} \left(\frac{n}{2} - 1, \frac{y^T y - y^T X (X^T X + \lambda^{(i)} J)^{-1} X^T y}{2} \right).$$

3. Using $\lambda^{(i)}$ and $\sigma^{(i)}$, we can generate $\beta^{(i)}$ from the multivariate normal distribution:

$$\beta^{(i)} \sim N \left(\hat{\beta}_{\text{ridge}}(\lambda^{(i)}), (\sigma^{(i)})^2 (X^T X + \lambda^{(i)} J)^{-1} \right)$$

Given posterior samples $\beta^{(i)}, \sigma^{(i)}, \lambda^{(i)}, i = 1, \dots, N$, we can plot the fitted values:

$$\beta_0^{(i)} + \beta_1^{(i)}(t-1) + \beta_2^{(i)} \text{ReLU}(t-2) + \dots + \beta_{n-1}^{(i)} \text{ReLU}(t-(n-1))$$

for $t = 1, \dots, n$. These fitted values give a general idea of the uncertainty in the β 's.

3 Likelihood vs Marginal Likelihood

The likelihood in this high-dimensional linear regression model is given by (10):

$$f_{\text{data}|\beta,\sigma}(\text{data}) \propto \frac{1}{\sigma^n} \exp\left(-\frac{\|y - X\beta\|^2}{2\sigma^2}\right).$$

Maximizing this likelihood leads to overfitting (the maximizing β will be the least squares estimate, and σ will be 0).

Under the prior (7) along with $\log \sigma \sim \text{unif}(-C, C)$, we can integrate the above likelihood with respect to this prior to get a marginal likelihood of the data in terms of λ alone:

$$f_{\text{data}|\lambda}(\text{data}) = \int f_{\text{data}|\lambda,\beta,\sigma}(\text{data}) f_{\beta|\sigma,\lambda}(\beta, \sigma) d\beta d\sigma.$$

The calculations from the previous section can be followed to show that:

$$f_{\text{data}|\lambda}(\text{data}) \propto \frac{\lambda^{(n-2)/2} |X^T X + \lambda J|^{-1/2}}{(y^T y - y^T X (X^T X + \lambda J)^{-1} X^T y)^{(n/2)-1}}.$$

The marginal likelihood provides a way of choosing λ that balances overfitting and underfitting. Recall that very small λ leads to fitted values that nearly interpolate the data, while very large λ leads to fitted values close to the least squares line. Unless the data are well-explained by a single line, we would like to choose λ between these extremes.

To understand how marginal likelihood addresses this tradeoff, consider its integral representation:

$$m(\lambda) := f_{\text{data}|\lambda}(\text{data}) \propto \int_0^\infty \left[\int f_{\text{data}|\beta,\sigma}(\text{data}) f_{\beta|\sigma,\lambda}(\beta) d\beta \right] \frac{d\sigma}{\sigma}. \quad (15)$$

Here we have used the prior $f_\sigma(\sigma) \propto 1/\sigma$. Because some of our priors are improper, $m(\lambda)$ is understood as an integrated likelihood defined up to a common multiplicative constant independent of λ .

First hold σ fixed and suppose λ is very large. Then the prior variances σ^2/λ of $\beta_2, \dots, \beta_{n-1}$ are small, so the prior concentrates on coefficients whose fitted values are close to a single line. If the data exhibit substantial departures from a line relative to the noise level σ , these coefficients have small likelihood. Consequently, concentrating the prior on them reduces the inner integral in (15).

On the other hand, suppose λ is very small, again holding σ fixed. The prior on $\beta_2, \dots, \beta_{n-1}$ is then spread over a very large region. Although some coefficients in this region fit the data extremely well, only a small fraction of the prior mass lies near these coefficients. In particular, the prior density contains the normalizing factor $\lambda^{(n-2)/2}$, which becomes small as λ decreases. Thus the ability to achieve an excellent fit does not by itself ensure a large integrated likelihood.

These competing effects explain how marginal likelihood balances fit and flexibility: it considers both the likelihood of parameter values and the prior weight assigned to them. Because σ is also integrated out, however, this argument does not guarantee that the maximizing λ lies away from the endpoints, or that overfitting and underfitting are always avoided.

For full Bayesian inference, we combine $m(\lambda)$ with the prior on λ . Our prior is uniform in $\log \lambda$, rather than in λ , so

$$f_{\log \lambda|\text{data}}(x) \propto m(e^x), \quad \log \lambda_{\min} \leq x \leq \log \lambda_{\max}.$$

Thus the shape of the posterior density on the log scale is determined by the marginal likelihood.

Note that this approach differs from selecting λ by cross-validation. Cross-validation evaluates predictive performance using training and validation splits, whereas the marginal-likelihood calculation uses the entire dataset without splitting it. For further discussion of Bayesian hyperparameter selection, cross-validation, and the intuition behind marginal likelihood, see http://www.inference.org.uk/mackay/Bayes_FAQ.html#gcv or <https://statmodeling.stat.columbia.edu/2011/12/04/david-mackay-and-occams-razor/>.